5. Rowhammer

(Based on materials from Onur Mutlu – ETH Zurich)

Arran Stewart

How Reliable/Secure/Safe is This Bridge?

Source: http://www.technologystudent.com/struct1/tacom1.png

Collapse of the Tacoma Narrows bridge

ource: http://www.wsdot.wa.gov/tnbhistory/connections/connections3.htm

How Secure Are These People?

Security is about preventing unforeseen consequences

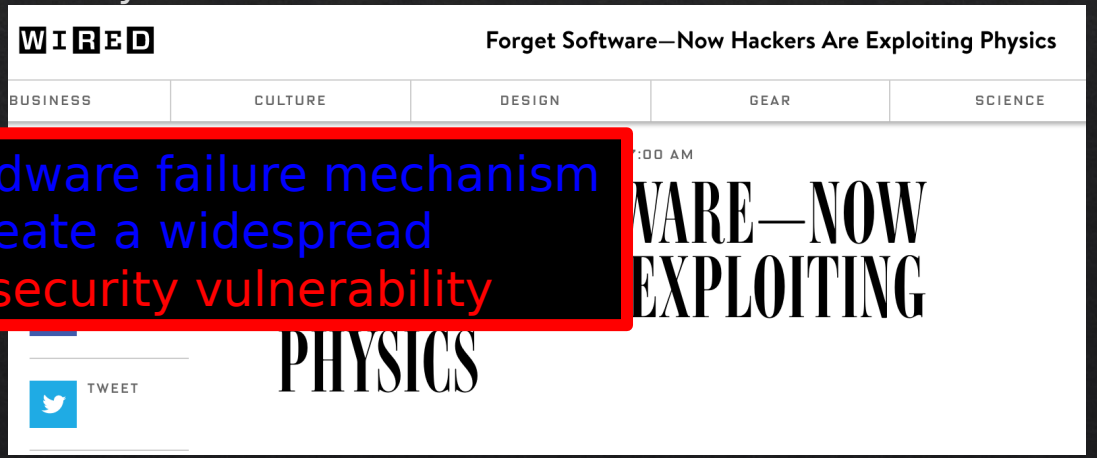
Source: https://s-media-cache-ak0.pinimg.com/originals/48/09/54/4809543a9c7700246a0cf8acdae27abf.jpg

How Safe & Secure Are Our Platforms?

Source: https://taxistartup.com/wp-content/uploads/2015/03/UK-Self-Driving-Cars.jpg

What Is RowHammer?

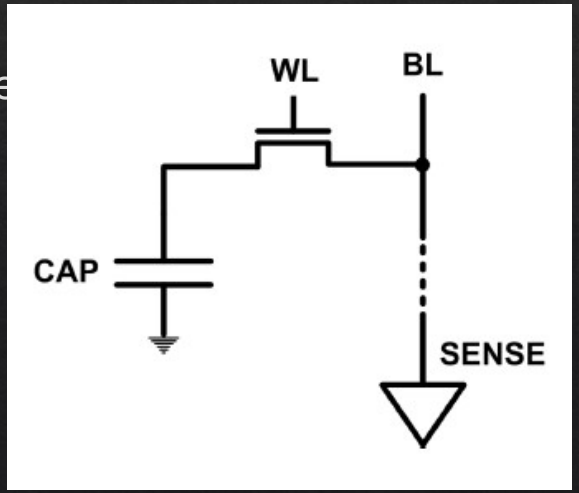
- One can predictably induce bit flips in commodity DRAM chips
  - >80% of the tested DRAM chips were vulnerable
  -
- First example of how a simple hardware failure mechanism can create a widespread system security vulnerability



A simple hardware failure mechanism can create a widespread system security vulnerability

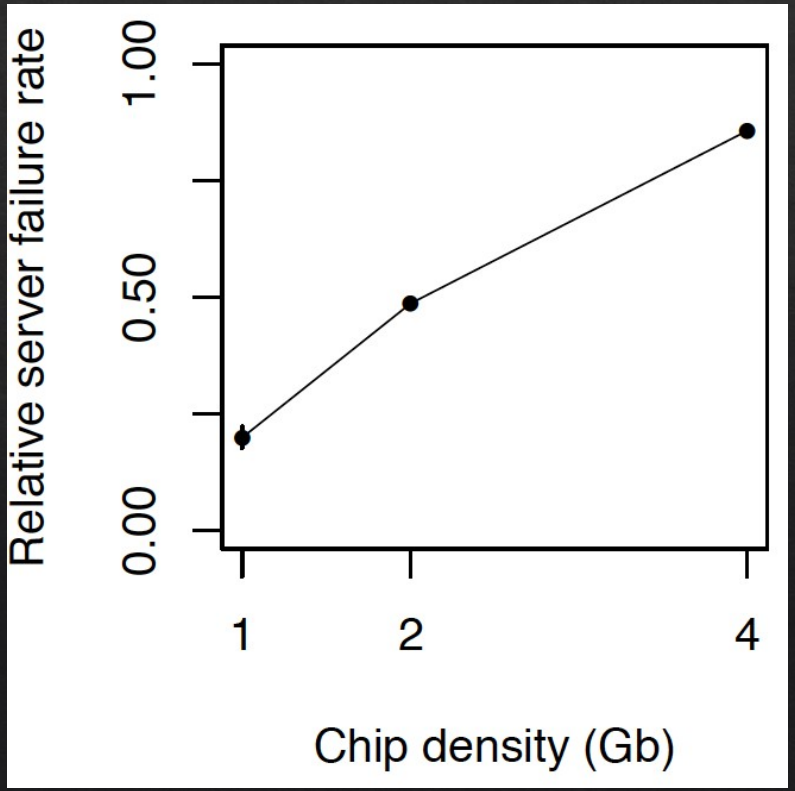
The DRAM Scaling Problem

- DRAM stores charge in a capacitor (charge-based memory)
  - Capacitor must be large enough for reliable sensing
  - Access transistor should be large enough for low leakage and high retention time
  - Scaling beyond 40-35nm (2013) is challenging [ITRS, 2009]
- DRAM capacity, cost, and energy/power hard to scale

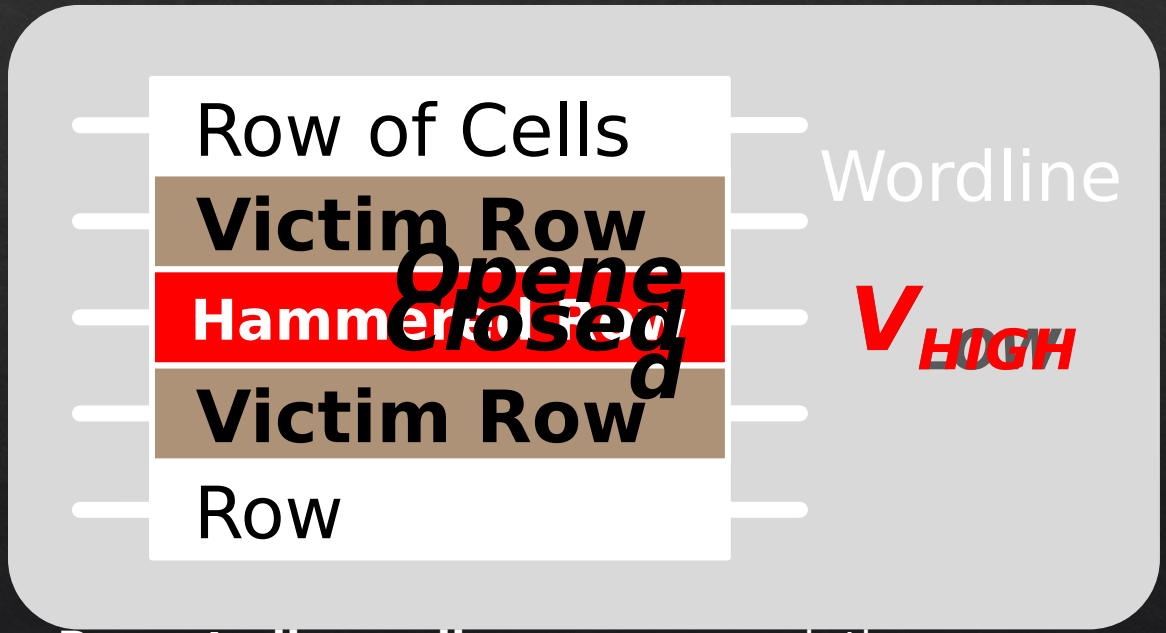


As Memory Scales, It Becomes Unreliable

- Data from all of Facebook's servers worldwide
  - Meza+, "Revisiting Memory Errors in Large-Scale Production Data Centers," DSN'15.



DRAM is Prone to Disturbance Errors



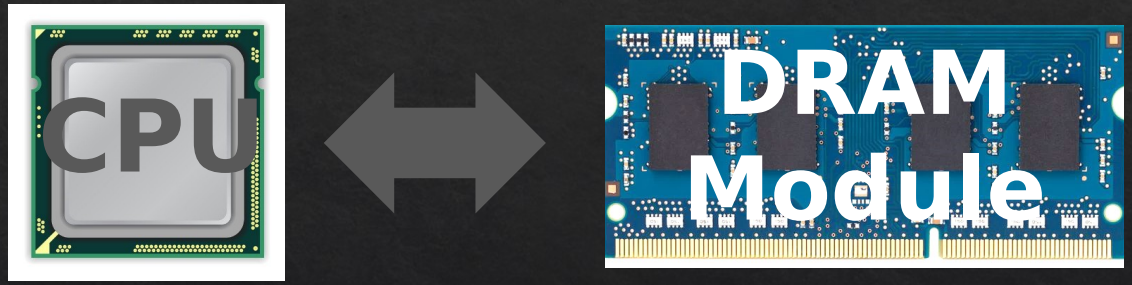
Repeatedly reading a row enough times (before memory gets refreshed) induces **disturbance errors** in **adjacent rows** in many **DRAM chips**

- More densely packed DRAM is more vulnerable
- DRAM with less error checking is more vulnerable

Why Does This Happen?

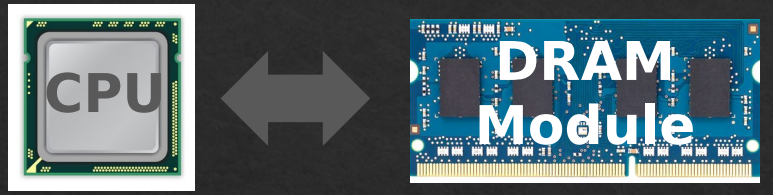
- DRAM cells are too close to each other!
  - They are not electrically isolated from each other
- Access to one cell affects the value in nearby cells
  - due to electrical interference between
    - the cells
    - wires used for accessing the cells
  - Also called cell-to-cell coupling/interference
- Example: When we activate (apply high voltage) to a row, an adjacent row gets slightly activated as well
  - Vulnerable cells in that slightly-activated row lose a little bit of charge
  - If RowHammer happens enough times, charge in such cells gets drained

A Simple Program Can Induce Many Errors

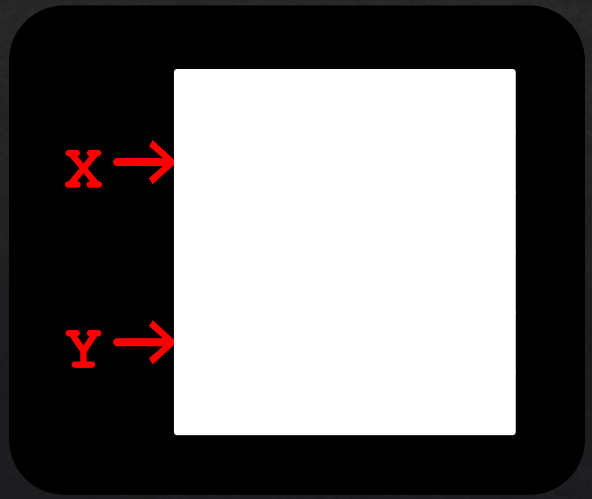


Download from: https://github.com/CMU-SAFARI/rowhammer

A Simple Program Can Induce Many Errors




```
loop:
  mov (X), %eax
  mov (Y), %ebx
  clflush (X)
  clflush (Y)
  mfence
  jmp loop
```



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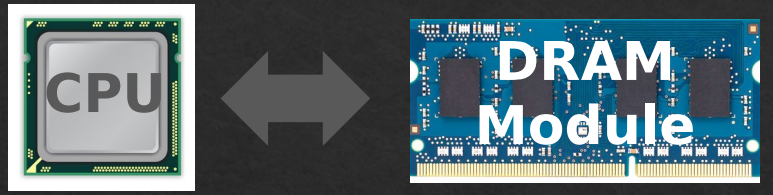
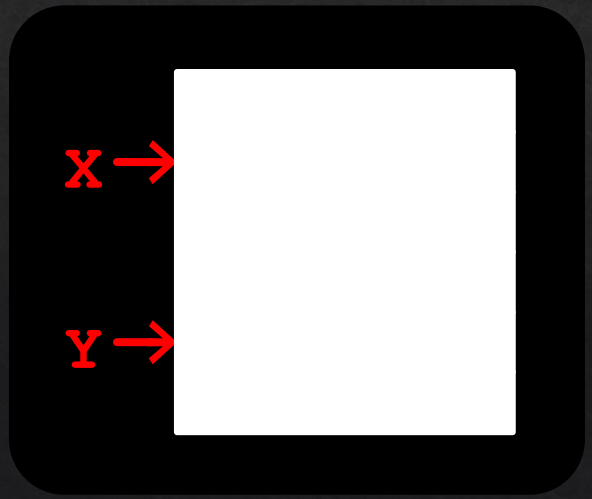
A Simple Program Can Induce Many Errors



1. Avoid ***cache hits***
   - Flush **X** from cache

2. Avoid ***row hits*** to **X**
   - Read **Y** in another row



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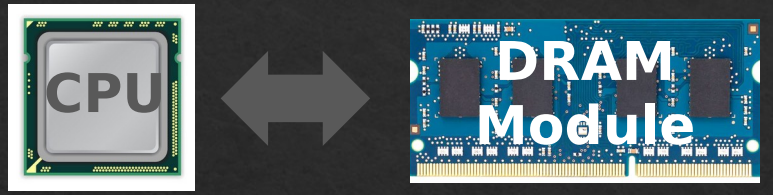


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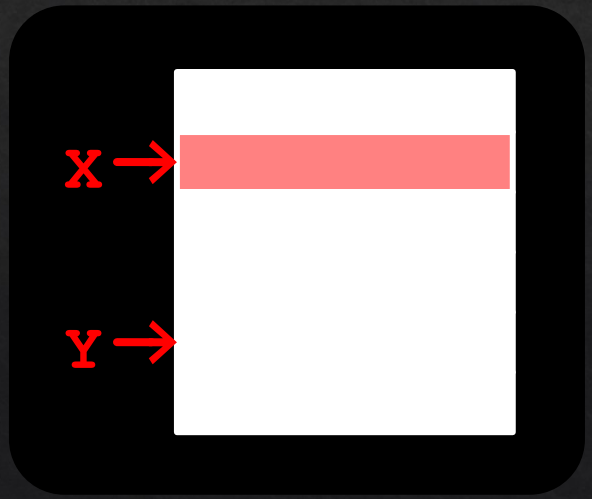

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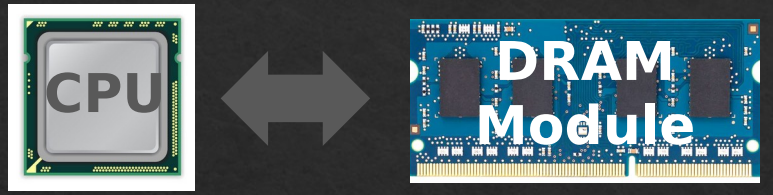


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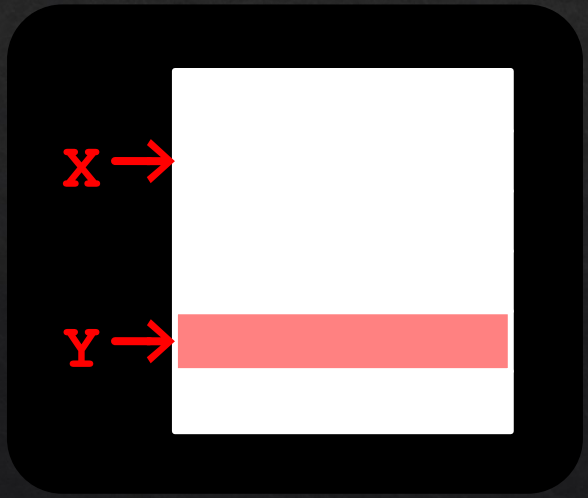

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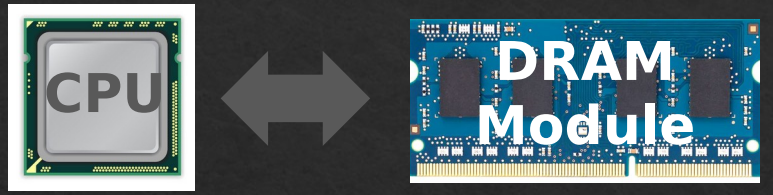


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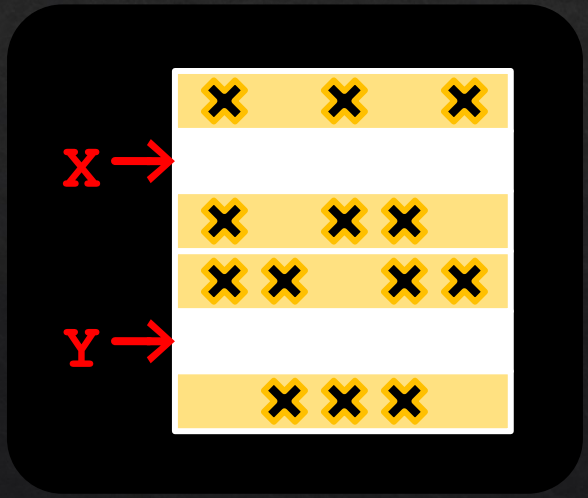


Download from: https://github.com/CMU-SAFARI/rowhammer

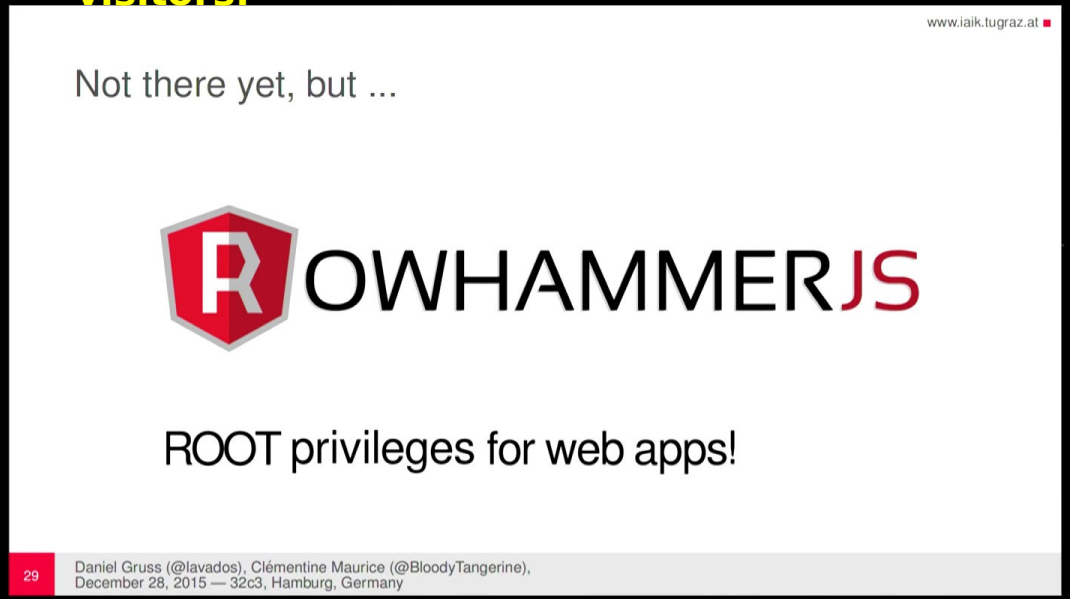
Observed Errors in Real Systems

CPU Architecture	Errors	Access-Rate
Intel Haswell (2013)	22.9K	12.3M/sec
Intel Ivy Bridge (2012)	20.7K	11.7M/sec
Intel Sandy Bridge (2011)	16.1K	11.6M/sec
AMD Piledriver (2012)	59	6.1M/sec

Security Implications

More Security Implications (I)

“We can gain unrestricted access to systems of website visitors.”



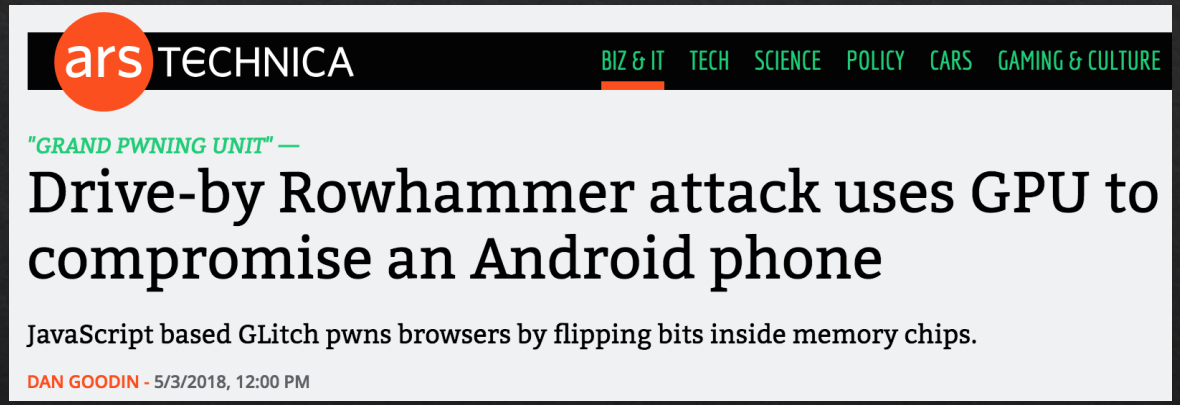
Rowhammer.js: A Remote Software-Induced Fault Attack in JavaScript (DIMVA’16)

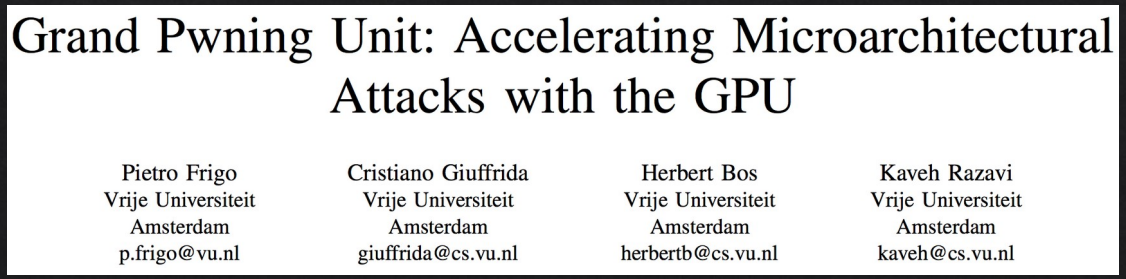
More Security Implications (II)

Drammer: Deterministic Rowhammer Attacks on Mobile Platforms, CCS’16

More Security Implications (III)

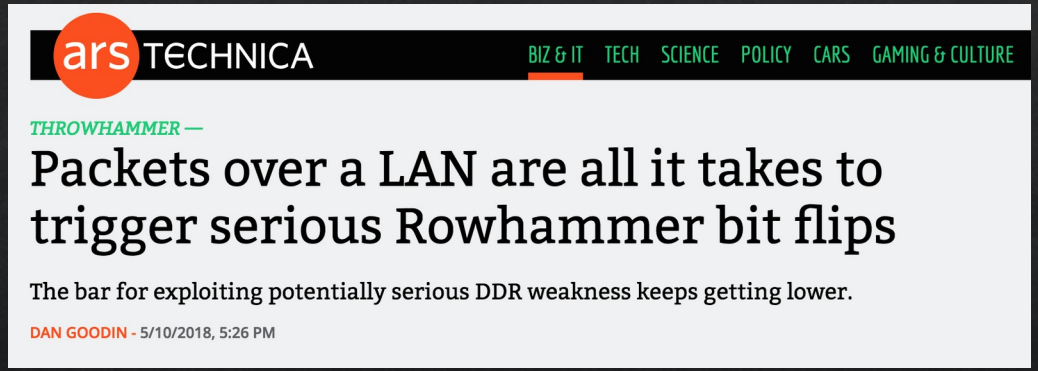
- Using an integrated GPU in a mobile system to remotely escalate privilege via the WebGL interface - IEEE S&P 2018





More Security Implications (IV)

- Rowhammer over RDMA (I) - USENIX ATC 2018



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THROWHAMMER —

Packets over a LAN are all it takes to trigger serious Rowhammer bit flips

The bar for exploiting potentially serious DDR weakness keeps getting lower.

DAN GOODIN - 5/10/2018, 5:26 PM

Throwhammer: Rowhammer Attacks over the Network and Defenses

Andrei Tatar
VU Amsterdam

Radhesh Krishnan
VU Amsterdam

Elias Athanasopoulos
University of Cyprus

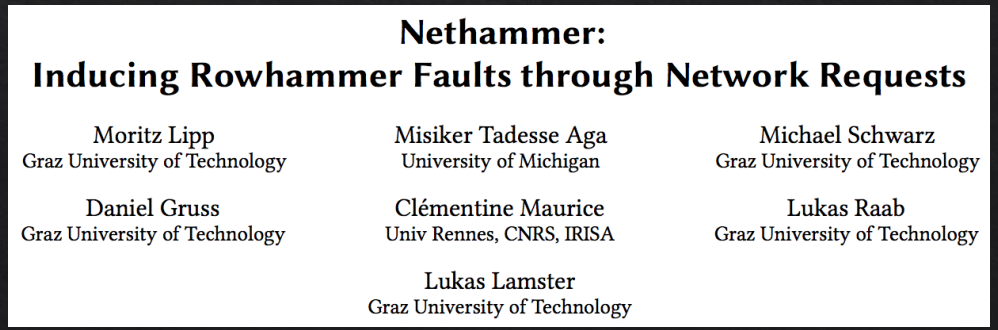
Cristiano Giuffrida
VU Amsterdam

Herbert Bos
VU Amsterdam

Kaveh Razavi
VU Amsterdam

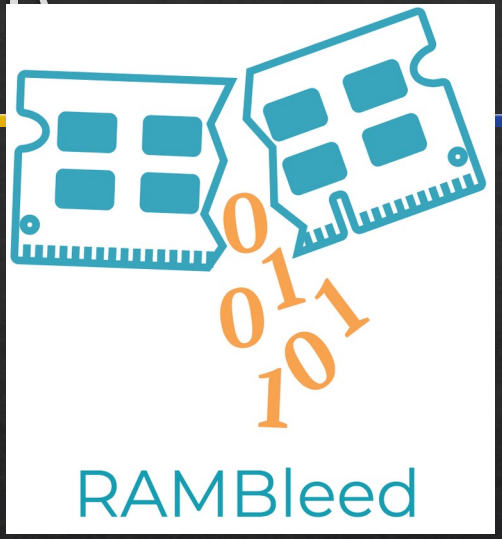
More Security Implications (V)

- Rowhammer over RDMA (II)



More Security Implications (V)

- IEEE S&P 2020



RAMBleed: Reading Bits in Memory Without Accessing Them

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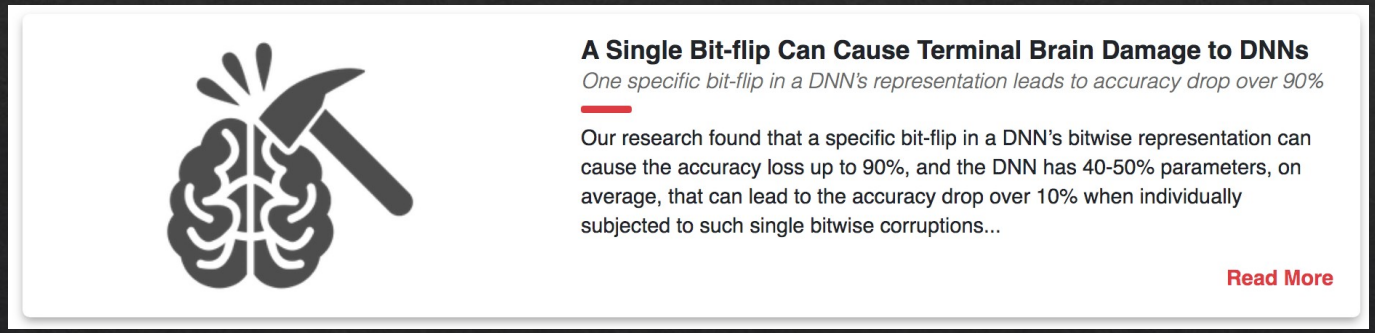
RAMBleed

- Attacker allocates memory in a controlled location.
- Rowhammer-like hammering is used induce probabilistic bit flips in rows *adjacent to* secret data (e.g. encryption keys)
- Attackers monitor which of their own memory cells flip (and how often), and deduce the original value of the victim’s bits.
- Over many iterations, this lets allows attacker to reconstruct sensitive data without ever directly reading it.

More Security Implications (VII)

- USENIX Security 2019



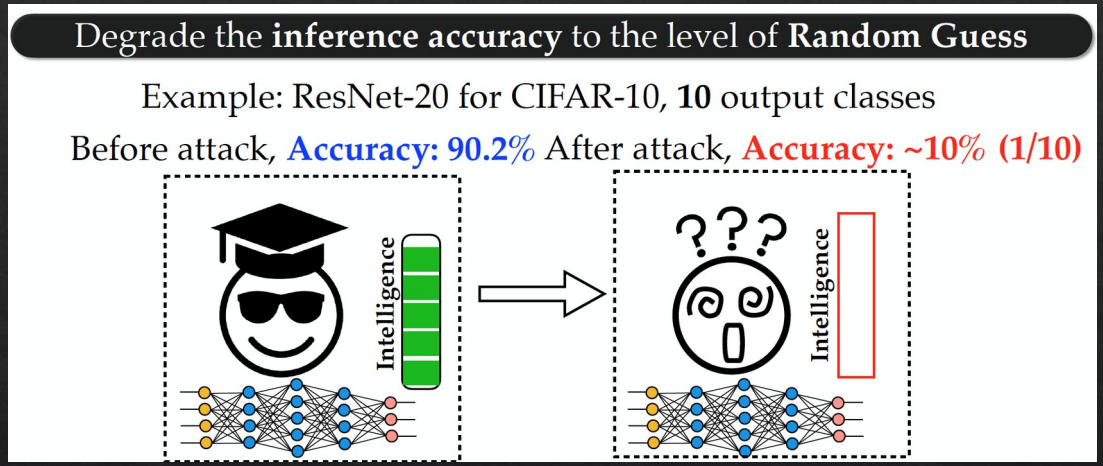
Terminal Brain Damage: Exposing the Graceless Degradation in Deep Neural Networks Under Hardware Fault Attacks

Sanghyun Hong, Pietro Frigo†, Yiğitcan Kaya, Cristiano Giuffrida†, Tudor Dumitraş

University of Maryland, College Park
†*Vrije Universiteit Amsterdam*

More Security Implications (VIII)

- USENIX Security 2020



DeepHammer: Depleting the Intelligence of Deep Neural Networks through Targeted Chain of Bit Flips

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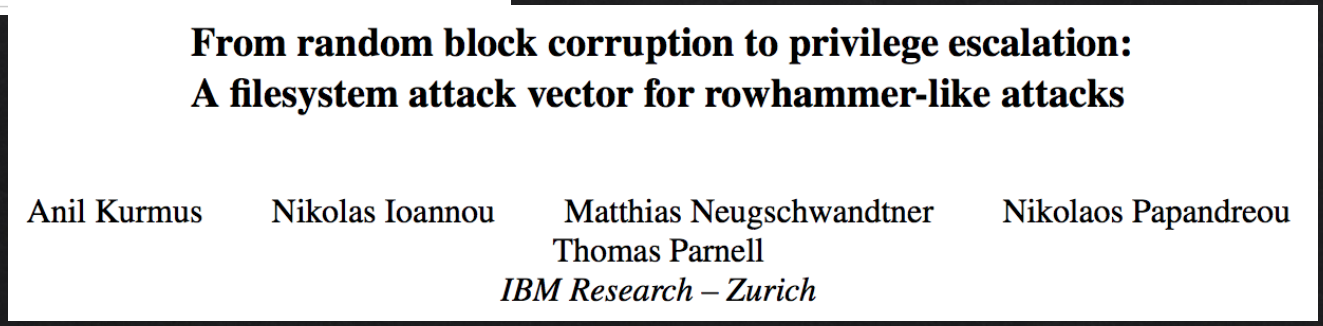
More Security Implications (IX)

- Rowhammer on MLC NAND Flash (based on [Cai+, HPCA 2017])



Tested DRAM Modules from 2008-2014

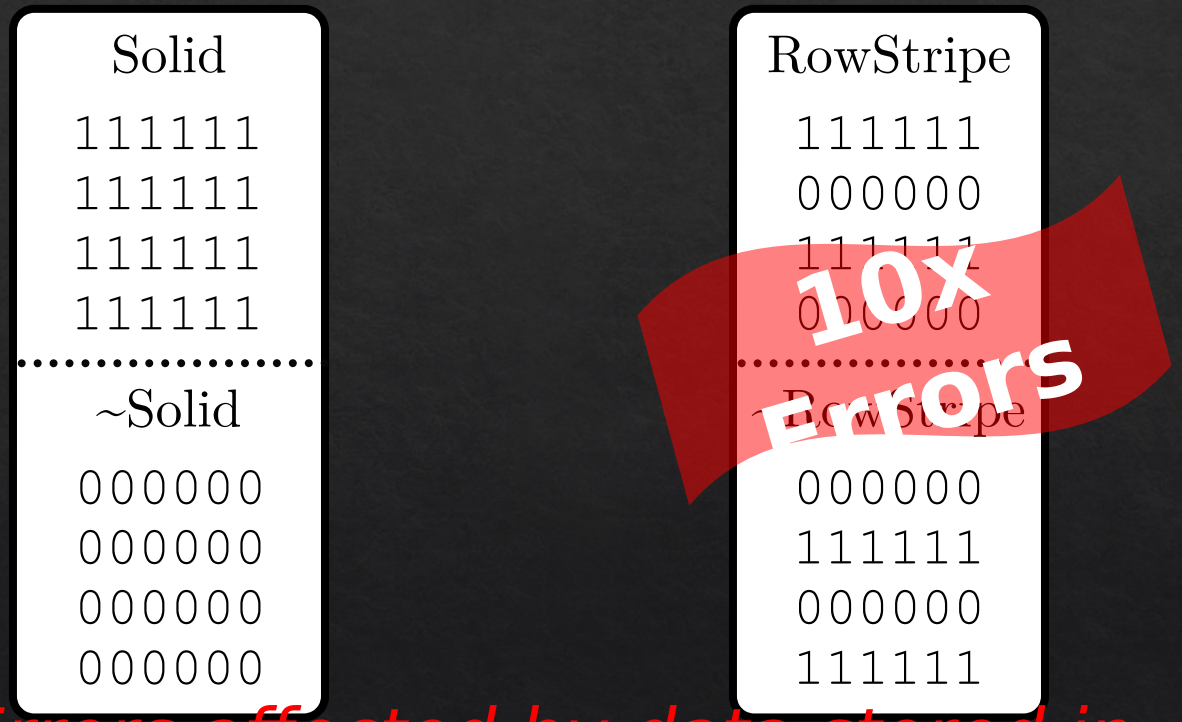
(129 total)

Manufacturer	Module	Date*	Timing†		Organization		Chip			Victims-per-Module			RI_{th} (ms)
		(yy-ww)	Freq (MT/s)	t_{RC} (ns)	Size (GB)	Chips	Size (Gb)‡	Pins	DieVersion§	Average	Minimum	Maximum	Min
A Total of 43 Modules	A_1	10-08	1066	50.625	0.5	4	1	×16	$\mathcal{B}$	0	0	0	–
	A_2	10-20	1066	50.625	1	8	1	×8	$\mathcal{F}$	0	0	0	–
	$A_{3\text{-}5}$	10-20	1066	50.625	0.5	4	1	×16	$\mathcal{B}$	0	0	0	–
	$A_{6\text{-}7}$	11-24	1066	49.125	1	4	2	×16	$\mathcal{D}$	7.8×10^1	5.2×10^1	1.0×10^2	21.3
	$A_{8\text{-}12}$	11-26	1066	49.125	1	4	2	×16	$\mathcal{D}$	2.4×10^2	5.4×10^1	4.4×10^2	16.4
	$A_{13\text{-}14}$	11-50	1066	49.125	1	4	2	×16	$\mathcal{D}$	8.8×10^1	1.7×10^1	1.6×10^2	26.2
	$A_{15\text{-}16}$	12-22	1600	50.625	1	4	2	×16	$\mathcal{D}$	9.5	9	1.0×10^1	34.4
	$A_{17\text{-}18}$	12-26	1600	49.125	2	8	2	×8	$\mathcal{M}$	1.2×10^2	3.7×10^1	2.0×10^2	21.3
	$A_{19\text{-}30}$	12-40	1600	48.125	2	8	2	×8	$\mathcal{K}$	8.6×10^6	7.0×10^6	$\mathbf{1.0 \times 10^7}$	**8.2**
	$A_{31\text{-}34}$	13-02	1600	48.125	2	8	2	×8	–	1.8×10^6	1.0×10^6	3.5×10^6	11.5
	$A_{35\text{-}36}$	13-14	1600	48.125	2	8	2	×8	–	4.0×10^1	1.9×10^1	6.1×10^1	21.3
	$A_{37\text{-}38}$	13-20	1600	48.125	2	8	2	×8	$\mathcal{K}$	1.7×10^6	1.4×10^6	2.0×10^6	9.8
	$A_{39\text{-}40}$	13-28	1600	48.125	2	8	2	×8	$\mathcal{K}$	5.7×10^4	5.4×10^4	6.0×10^4	16.4
	A_{41}	14-04	1600	49.125	2	8	2	×8	–	2.7×10^5	2.7×10^5	2.7×10^5	18.0
	$A_{42\text{-}43}$	14-04	1600	48.125	2	8	2	×8	$\mathcal{K}$	0.5	0	1	62.3
B Total of 54 Modules	B_1	08-49	1066	50.625	1	8	1	×8	$\mathcal{D}$	0	0	0	–
	B_2	09-49	1066	50.625	1	8	1	×8	$\mathcal{E}$	0	0	0	–
	B_3	10-19	1066	50.625	1	8	1	×8	$\mathcal{F}$	0	0	0	–
	B_4	10-31	1333	49.125	2	8	2	×8	$\mathcal{C}$	0	0	0	–
	B_5	11-13	1333	49.125	2	8	2	×8	$\mathcal{C}$	0	0	0	–
	B_6	11-16	1066	50.625	1	8	1	×8	$\mathcal{F}$	0	0	0	–
	B_7	11-19	1066	50.625	1	8	1	×8	$\mathcal{F}$	0	0	0	–
	B_8	11-25	1333	49.125	2	8	2	×8	$\mathcal{C}$	0	0	0	–
	B_9	11-37	1333	49.125	2	8	2	×8	$\mathcal{D}$	1.9×10^6	1.9×10^6	1.9×10^6	11.5
	$B_{10\text{-}12}$	11-46	1333	49.125	2	8	2	×8	$\mathcal{D}$	2.2×10^6	1.5×10^6	$\mathbf{2.7 \times 10^6}$	11.5
	B_{13}	11-49	1333	49.125	2	8	2	×8	$\mathcal{C}$	0	0	0	–
	B_{14}	12-01	1866	47.125	2	8	2	×8	$\mathcal{D}$	9.1×10^5	9.1×10^5	9.1×10^5	**9.8**
	$B_{15\text{-}31}$	12-10	1866	47.125	2	8	2	×8	$\mathcal{D}$	9.8×10^5	7.8×10^5	1.2×10^6	11.5
	B_{32}	12-25	1600	48.125	2	8	2	×8	$\mathcal{E}$	7.4×10^5	7.4×10^5	7.4×10^5	11.5
	$B_{33\text{-}42}$	12-28	1600	48.125	2	8	2	×8	$\mathcal{E}$	5.2×10^5	1.9×10^5	7.3×10^5	11.5
	$B_{43\text{-}47}$	12-31	1600	48.125	2	8	2	×8	$\mathcal{E}$	4.0×10^5	2.9×10^5	5.5×10^5	13.1
	$B_{48\text{-}51}$	13-19	1600	48.125	2	8	2	×8	$\mathcal{E}$	1.1×10^5	7.4×10^4	1.4×10^5	14.7
	$B_{52\text{-}53}$	13-40	1333	49.125	2	8	2	×8	$\mathcal{D}$	2.6×10^4	2.3×10^4	2.9×10^4	21.3
	B_{54}	14-07	1333	49.125	2	8	2	×8	$\mathcal{D}$	7.5×10^3	7.5×10^3	7.5×10^3	26.2
C Total of 32 Modules	C_1	10-18	1333	49.125	2	8	2	×8	$\mathcal{A}$	0	0	0	–
	C_2	10-20	1066	50.625	2	8	2	×8	$\mathcal{A}$	0	0	0	–
	C_3	10-22	1066	50.625	2	8	2	×8	$\mathcal{A}$	0	0	0	–
	$C_{4\text{-}5}$	10-26	1333	49.125	2	8	2	×8	$\mathcal{B}$	8.9×10^2	6.0×10^2	1.2×10^3	29.5
	C_6	10-43	1333	49.125	1	8	1	×8	$\mathcal{T}$	0	0	0	–
	C_7	10-51	1333	49.125	2	8	2	×8	$\mathcal{B}$	4.0×10^2	4.0×10^2	4.0×10^2	29.5
	C_8	11-12	1333	46.25	2	8	2	×8	$\mathcal{B}$	6.9×10^2	6.9×10^2	6.9×10^2	21.3
	C_9	11-19	1333	46.25	2	8	2	×8	$\mathcal{B}$	9.2×10^2	9.2×10^2	9.2×10^2	27.9
	C_{10}	11-31	1333	49.125	2	8	2	×8	$\mathcal{B}$	3	3	3	39.3
	C_{11}	11-42	1333	49.125	2	8	2	×8	$\mathcal{B}$	1.6×10^2	1.6×10^2	1.6×10^2	39.3
	C_{12}	11-48	1600	48.125	2	8	2	×8	$\mathcal{C}$	7.1×10^4	7.1×10^4	7.1×10^4	19.7
	C_{13}	12-08	1333	49.125	2	8	2	×8	$\mathcal{C}$	3.9×10^4	3.9×10^4	3.9×10^4	21.3
	$C_{14\text{-}15}$	12-12	1333	49.125	2	8	2	×8	$\mathcal{C}$	3.7×10^4	2.1×10^4	5.4×10^4	21.3
	$C_{16\text{-}18}$	12-20	1600	48.125	2	8	2	×8	$\mathcal{C}$	3.5×10^3	1.2×10^3	7.0×10^3	27.9
	C_{19}	12-23	1600	48.125	2	8	2	×8	$\mathcal{E}$	1.4×10^5	1.4×10^5	1.4×10^5	18.0
	C_{20}	12-24	1600	48.125	2	8	2	×8	$\mathcal{C}$	6.5×10^4	6.5×10^4	6.5×10^4	21.3
	C_{21}	12-26	1600	48.125	2	8	2	×8	$\mathcal{C}$	2.3×10^4	2.3×10^4	2.3×10^4	24.6
	C_{22}	12-32	1600	48.125	2	8	2	×8	$\mathcal{C}$	1.7×10^4	1.7×10^4	1.7×10^4	22.9
	$C_{23\text{-}24}$	12-37	1600	48.125	2	8	2	×8	$\mathcal{C}$	2.3×10^4	1.1×10^4	3.4×10^4	18.0
	$C_{25\text{-}30}$	12-41	1600	48.125	2	8	2	×8	$\mathcal{C}$	2.0×10^4	1.1×10^4	3.2×10^4	19.7
	C_{31}	13-11	1600	48.125	2	8	2	×8	$\mathcal{C}$	3.3×10^5	3.3×10^5	$\mathbf{3.3 \times 10^5}$	**14.7**
	C_{32}	13-35	1600	48.125	2	8	2	×8	$\mathcal{C}$	3.7×10^4	3.7×10^4	3.7×10^4	21.3

∗ We report the manufacture date marked on the chip packages, which is more accurate than other dates that can be gleaned from a module.
† We report timing constraints stored in the module's on-board ROM [33], which is read by the system BIOS to calibrate the memory controller.
‡ The maximum DRAM chip size supported by our testing platform is 2Gb.
§ We report DRAM die versions marked on the chip packages, which typically progress in the following manner: $\mathcal{M} \rightarrow \mathcal{A} \rightarrow \mathcal{B} \rightarrow \mathcal{C} \rightarrow \cdots$.

Table 3. Sample population of 129 DDR3 DRAM modules, categorized by manufacturer and sorted by manufacture date

❸ Data Pattern



Errors affected by data stored in other cells

Other Key Observations

- *Victim Cells $\neq$ Retention-Weak Cells*
  - Almost no overlap between them
- *Errors are repeatable*
  - Across ten iterations of testing, >70% of victim cells had errors in every iteration
- *As many as 4 errors per cache-line*
  - Simple ECC (e.g., SECDED) cannot prevent all errors
- *Cells affected by two aggressors on either side*
  - Double sided hammering

RowHammer continues to be relevant

Revisiting RowHammer: An Experimental Analysis of Modern DRAM Devices and Mitigation Techniques

Jeremie S. Kim[§†] Minesh Patel[§] A. Giray Yağlıkçı[§]
Hasan Hassan[§] Roknoddin Azizi[§] Lois Orosa[§] Onur Mutlu[§†]
[§]*ETH Zürich* [†]*Carnegie Mellon University*

(2020)

Understanding RowHammer Under Reduced Wordline Voltage: An Experimental Study Using Real DRAM Devices

A. Giray Yağlıkçı[1] Haocong Luo[1] Geraldo F. de Oliviera[1] Ataberk Olgun[1] Minesh Patel[1]
Jisung Park[1] Hasan Hassan[1] Jeremie S. Kim[1] Lois Orosa[1,2] Onur Mutlu[1]
[1]*ETH Zürich* [2]*Galicia Supercomputing Center (CESGA)*

(2022)

An Experimental Analysis of RowHammer in HBM2 DRAM Chips

Ataberk Olgun[1] Majd Osseiran[1,2] A. Giray Yağlıkçı[1] Yahya Can Tuğrul[1]
Haocong Luo[1] Steve Rhyner[1] Behzad Salami[1] Juan Gomez Luna[1] Onur Mutlu[1]
[1]*SAFARI Research Group, ETH Zürich* [2]*American University of Beirut*

(2023)

RowHammer Solutions

Naive Solutions

❶ *Throttle accesses to same row*

- Limit access-interval: $\geq 500\text{ns}$
- Limit number of accesses: $\leq 128\text{K}$ $(=64\text{ms}/500\text{ns})$

❷ *Refresh more frequently*

- Shorten refresh-interval by $\sim 7\text{x}$

Both naive solutions introduce significant overhead in performance and power

Two Types of RowHammer Solutions

- Immediate
  - To protect the vulnerable DRAM chips in the field
  - Limited possibilities
- Longer-term
  - To protect future DRAM chips
  - Wider range of protection mechanisms

Some Potential Solutions (ISCA 2014)

- Make better DRAM chips — Cost
- Refresh frequently — Power, Performance
- Sophisticated ECC — Cost, Power
- Access counters — Cost, Power, Complexity

Apple’s Security Patch for RowHammer

- https://support.apple.com/en-gb/HT204934

Available for: OS X Mountain Lion v10.8.5, OS X Mavericks v10.9.5

Impact: A malicious application may induce memory corruption to escalate privileges

Description: A disturbance error, also known as Rowhammer, exists with some DDR3 RAM that could have led to memory corruption. This issue was mitigated by increasing memory refresh rates.

CVE-ID

CVE-2015-3693 : Mark Seaborn and Thomas Dullien of Google, working from original research by Yoongu Kim et al (2014)

HP, Lenovo, and many other vendors released similar patches

RowHammer Solution: PARA

- **PARA:** *Probabilistic Adjacent Row Activation*
- **Key Idea**
  - **After closing a row, we activate (i.e., refresh) one of its neighbors with a low probability**: $p = 0.005$
- **Reliability Guarantee**
  - When $p=0.005$, errors in one year: 9.4×10^{-14}
  - By adjusting the value of p, we can vary the strength of protection against errors

Advantages of PARA

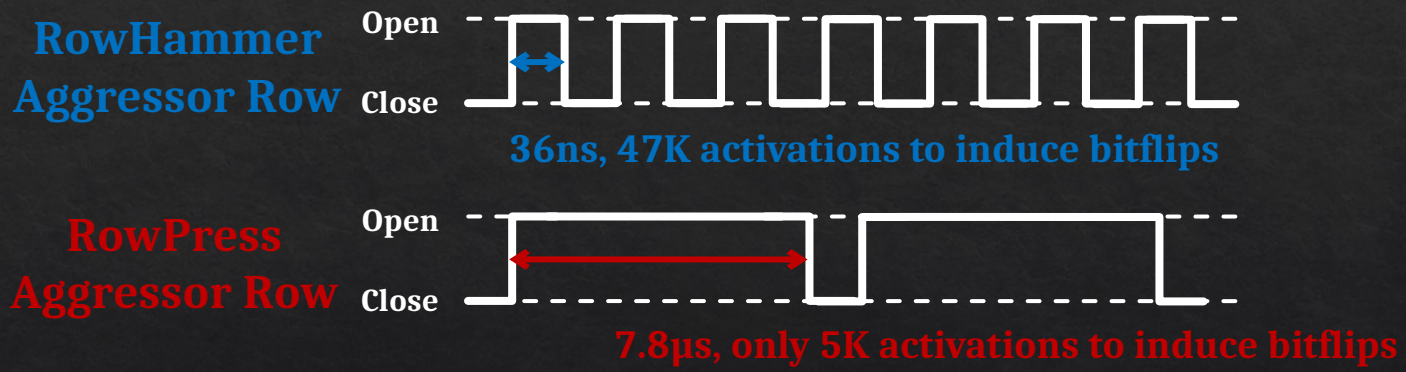
- *PARA refreshes rows infrequently*
  - Low power
  - Low performance-overhead
    - Average slowdown: 0.20% (for 29 benchmarks)
    - Maximum slowdown: 0.75%
- *PARA is stateless*
  - Low cost
  - Low complexity

- *PARA is an effective and low-overhead solution to prevent disturbance errors*

What next?

RowPress vs. RowHammer

Instead of using a high activation count,
☞ increase the time that the aggressor row stays open



Can cause bitflips even with **ONLY ONE activation**
in extreme cases where the row stays open for 30ms

Architecting Future Memory for Security

- **Understand**: Methods for vulnerability modeling & discovery
  - Modeling and prediction based on real (device) data and analysis
  - Understanding vulnerabilities
  - Developing reliable metrics
- **Architect**: Principled architectures with security as key concern
  - Good partitioning of duties across the stack
  - Cannot give up performance and efficiency
  - Patch-ability in the field
- **Design & Test**: Principled design, automation, (online) testing
  - Design for security
  - High coverage and good interaction with system reliability methods

Two Major Future RowHammer Directions

- **Understanding RowHammer**
  - Many effects still need to be rigorously examined
    - Aging of DRAM Chips
    - Environmental Conditions
    - Memory Access Patterns
    - Memory Controller & System Design Decisions
    - …
- **Solving RowHammer**
  - Flexible and Efficient RowHammer Solutions are necessary
    - In-field patchable / reconfigurable / programmable solutions
  - Co-architecting System and Memory is important
    - To avoid performance and denial-of-service problems

Google’s Original RowHammer Attack

Presented by Mark Seaborn and Thomas Dullien at BlackHat 2015

RowHammer Attack

Kernel exploit

- x86 page tables entries (PTEs) are **dense and trusted**
  - They control access to physical memory
  - A bit flip in a PTE's physical page number can give a process access to a different physical page
- Aim of exploit: Get access to a page table
  - Gives access to all of physical memory
- Maximise chances that a bit flip is useful:
  - Spray physical memory with page tables
  - Check for useful, repeatable bit flip first

RowHammer Attack

x86-64 Page Table Entries (PTEs)

- Page table is a 4k page containing array of 512 PTEs
- Each PTE is 64 bits, containing:

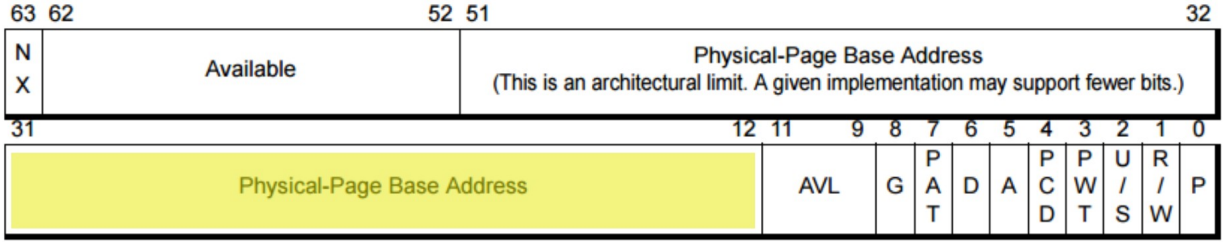
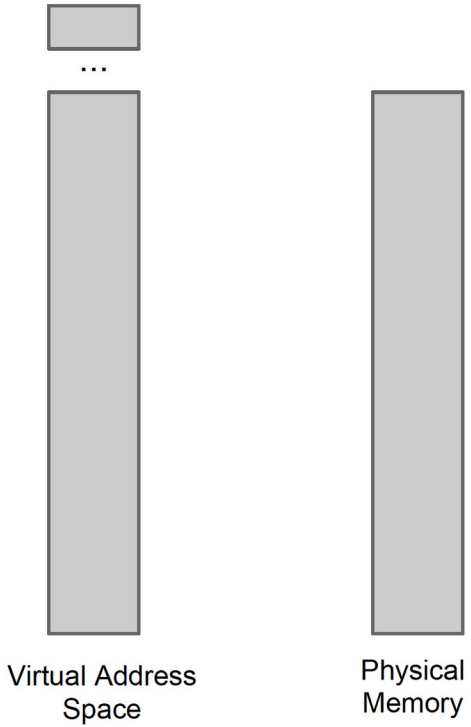
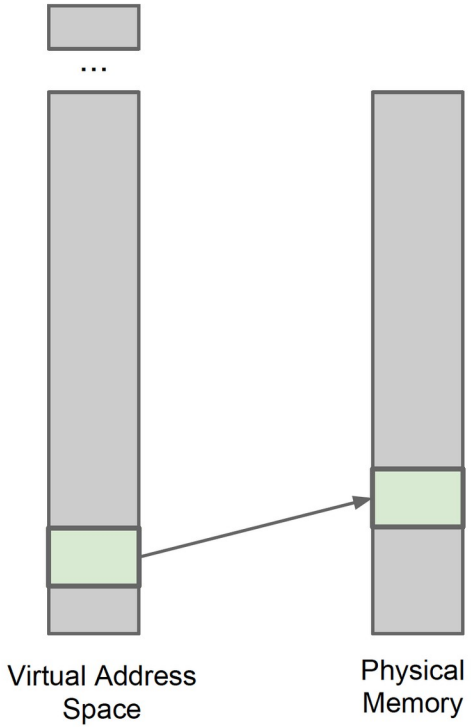


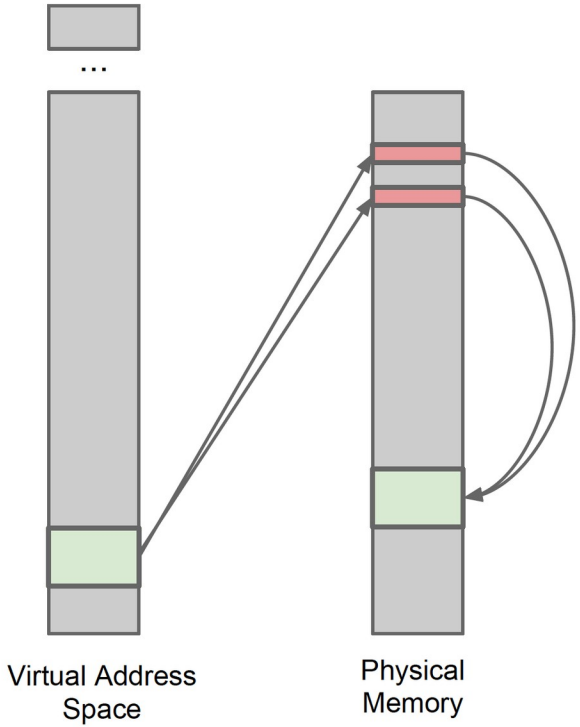
Figure 5-21. 4-Kbyte PTE—Long Mode

- Could flip:
  - "Writable" permission bit (RW): 1 bit → 2% chance
  - Physical page number: 20 bits on 4GB system → 31% chance

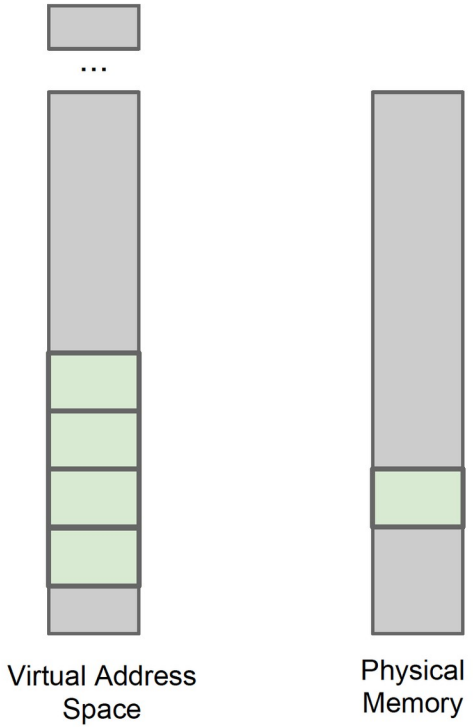

...
Virtual Address Space
Physical Memory



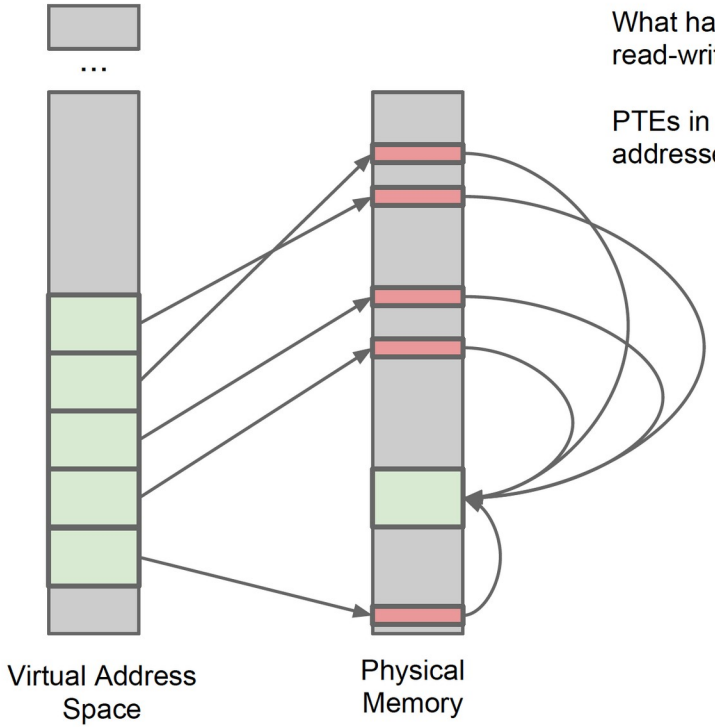
What happens when we map a file with read-write permissions?



What happens when we map a file with read-write permissions? Indirection via page tables.

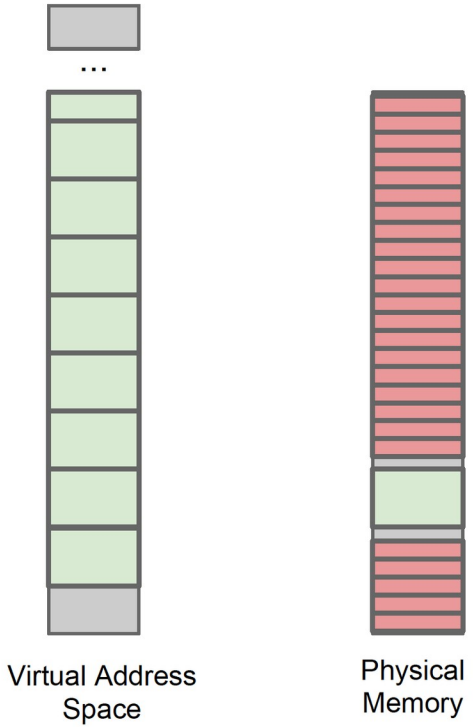


What happens when we repeatedly map a file with read-write permissions?



What happens when we repeatedly map a file with read-write permissions?

PTEs in physical memory help resolve virtual addresses to physical pages.



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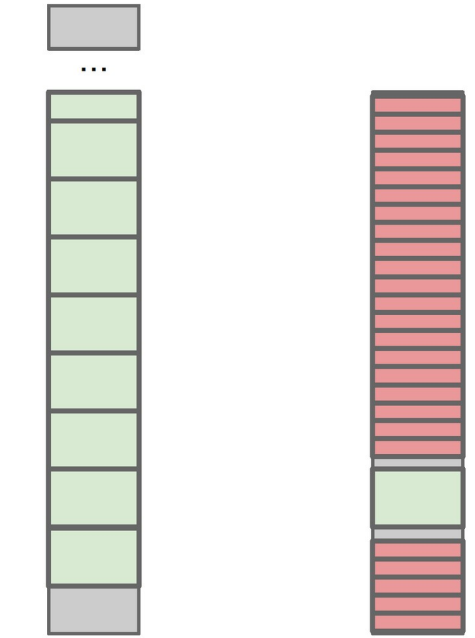
PTEs in physical memory help resolve virtual addresses to physical pages.

We can fill physical memory with PTEs.



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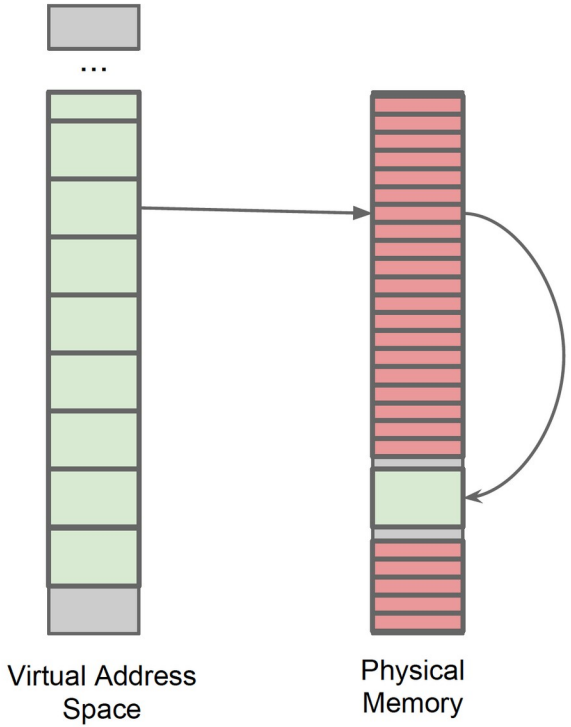
PTEs in physical memory help resolve virtual addresses to physical pages.

We can fill physical memory with PTEs.

Each of them points to pages in the same physical file mapping.



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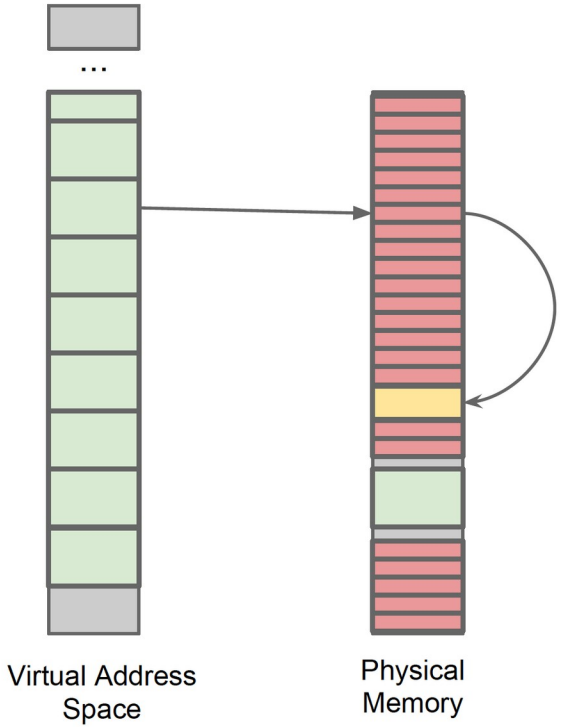

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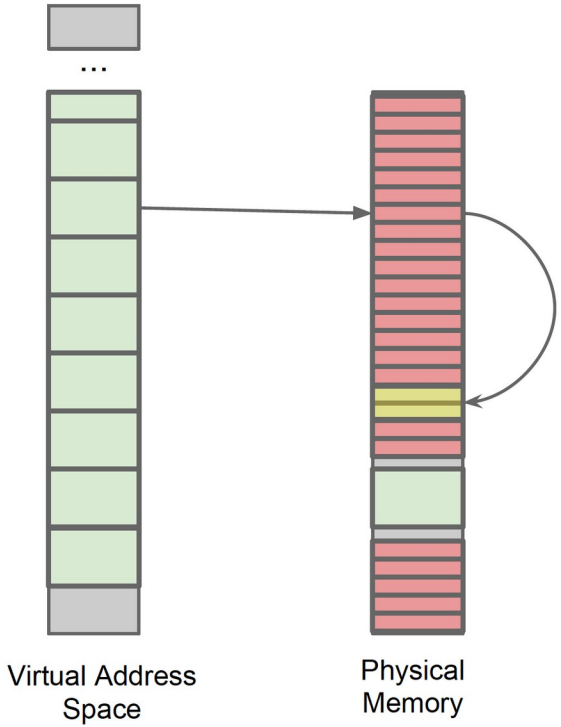
… the corresponding virtual address now points to a wrong physical page - with RW access.



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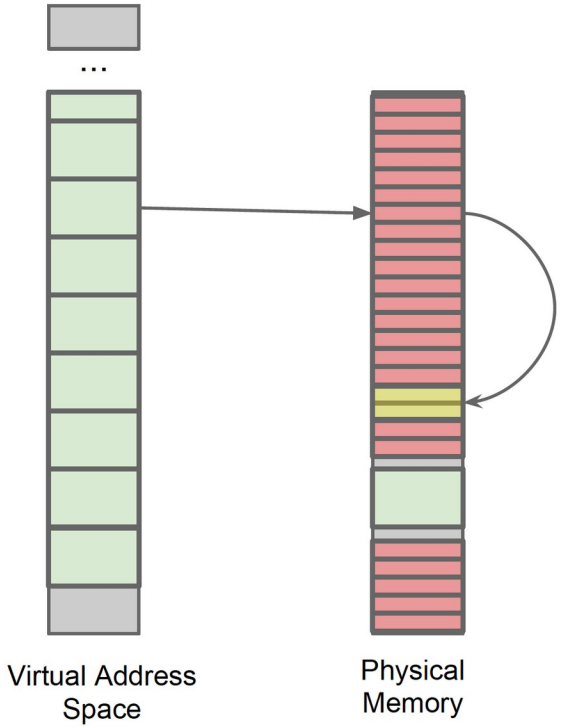
Chances are this wrong page contains a page table itself.



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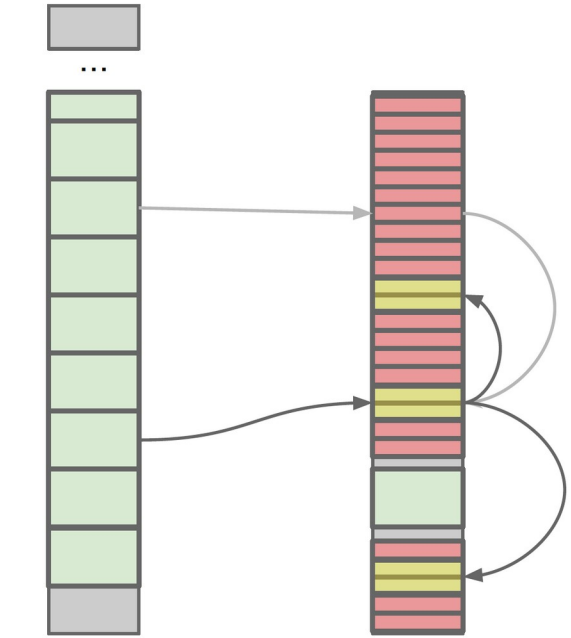
An attacker that can read / write page tables …



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Chances are this wrong page contains a page table itself.

An attacker that can read / write page tables can use that to map **any** memory read-write.

Exploit strategy

Privilege escalation in 7 easy steps …

1. Allocate a large chunk of memory
2. Search for locations prone to flipping
3. Check if they fall into the “right spot” in a PTE for allowing the exploit
4. Return that particular area of memory to the operating system
5. Force OS to re-use the memory for PTEs by allocating massive quantities of address space
6. Cause the bitflip - shift PTE to point into page table
7. Abuse R/W access to all of physical memory

In practice, there are many complications.

More info

- Computer Architecture, Fall 2021, Lecture 5
  - RowHammer (ETH Zürich, Fall 2021)
  - https://www.youtube.com/watch?v=7wVKnPj3NVw&list=PL5Q2soXY2Zi-Mnk1PxjEIG32HAGILkTOF&index=5
- Computer Architecture, Fall 2021, Lecture 6
  - RowHammer and Secure & Reliable Memory (ETH Zürich, Fall 2021)
  - https://www.youtube.com/watch?v=HNd4skQrt6I&list=PL5Q2soXY2Zi-Mnk1PxjEIG32HAGILkTOF&index=6
- Rowhammer attack from Google
  - https://www.blackhat.com/docs/us-15/materials/us-15-Seaborn-Exploiting-The-DRAM-Rowhammer-Bug-To-Gain-Kernel-Privileges.pdf
  - https://www.youtube.com/watch?v=0U7511Fb4to

References

- Materials from
  - Onur Mutlu, ETH Zurich